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When personalization deters delegation to algorithms: Experimental evidence

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Background

- Huge & unconnected literature on algorithm/automation/AI/… reliance (→ review by Chugunova & Sele, 2022)
- Most of the literature: setups with clearly identifiable mistakes
- But in many real-life settings, preferences determine what is a mistake
- E.g., robo-advisors: maximal payoff *vs* best fit to risk-parameters

Relevance?

Well-known (Swiss) robo-advisors(' marketing strategies)

- "We rely on a low-cost, passive investment strategy and have selected a total of 9 equity, bond and real estate ETFs from over 1,500 ETFs (exchange-traded funds) and put them together in such a way that they achieve an optimal risk/return ratio." (findependent.ch; for GER, see e.g., whitebox.eu)
- "Chat with Selma: Selma looks at your financial life and learns how you should structure your finances. Get your plan: Selma creates a personalized investment plan that fits your life. Sit back and relax! Selma takes care of your money from now on and does all the buying and selling for you." (selma.com; for GER, see e.g., quirion.de)

Research Question

- Does algorithm reliance differ for **objective optimization *vs.* preference-based personalisation** in a preferential-choice environment?
  (i) ...at first contact?
  (ii) ...overall?

Objective vs Personalised Algorithms

- What do we mean by **objective optimization**?
  - an algorithm that chooses some ‘objectively ideal’ option independently of a subject’s preferences (here: **highest EV**)
- What do we mean by **personalised**?
  - an algorithm that measures a subject’s preferences and chooses accordingly (here: **implementation of revealed preferences**)

Objective vs Personalised Algorithms

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  - an algorithm that measures a subject's preferences and chooses accordingly (here: **implementation of revealed preferences**)
- In the experiment, we describe both algorithms in intuitive terms:

EV: *'The algorithm works out how much each option would yield on average and selects the option for which this is highest.'*

Prefs: *'The algorithm analyses your choices from Part 1 and selects the best option for you in light of your Part-1 choices.'*

Experimental Design

Part 1 – Elicitation of preferences

- ordered lotteries, no foils, unlimited time
- "hypothetical, but the choices may be meaningful for Part 2"

Part 2 – Lottery Selection Task

- could stand for an investment task
- complexity (number of options, outcomes under different SOTW)
- time pressure (think: relative to the number of options)

Tasks

	State 1	State 2	State 3	State 4	State 5	State 6	State 7	State 8	State 9	State 10
Option 1	225	225	225	225	225	225	225	205	0	0
Option 2	525	525	525	525	525	0	0	0	0	0

State:	1	2	3	4	5	6	7	8	9	10
Option 1	225	225	0	225	0	225	225	225	225	0
Option 2	525	0	505	0	525	0	0	0	525	505
Option 3	525	0	525	0	525	0	0	0	525	525
Option 4	225	225	0	225	205	205	205	225	225	0
Option 5	225	205	0	205	205	225	225	205	225	0
Option 6	225	225	0	225	205	225	225	225	225	0
Option 7	525	0	525	0	525	0	0	0	225	525
Option 8	505	0	525	0	505	0	0	0	505	525

- For further distraction: 4 filler tasks in between

Tasks

	State 1	State 2	State 3	State 4	State 5	State 6	State 7	State 8	State 9	State 10
Option 1	225	225	225	225	225	225	225	205	0	0
Option 2	525	525	525	525	525	0	0	0	0	0

State:	1	2	3	4	5	6	7	8	9	10
Option 1	225	225	0	225	0	225	225	225	225	0
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Option 3	525	0	525	0	525	0	0	0	525	525
Option 4	225	225	0	225	205	205	205	225	225	0
Option 5	225	205	0	205	205	225	225	205	225	0
Option 6	225	225	0	225	205	225	225	225	225	0
Option 7	525	0	525	0	525	0	0	0	225	525
Option 8	505	0	525	0	505	0	0	0	505	525

- For further distraction: 4 filler tasks in between

Lottery-Selection Task

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- 8 rounds: Algorithm choice—10' viewing time—10' decision/waiting time
- P-Bet vs \$-Bet, e.g.: (110, 0.7; 0) vs (175, 0.3; 0)

Lottery-Selection Task

- 8 rounds: Algorithm choice—10' viewing time—10' decision/waiting time
- P-Bet vs $-Bet, e.g.: (110, 0.7; 0) vs (175, 0.3; 0)
- Generating complexity: 1. Inclusion of **dominated lotteries**

  → humans **should** and algorithms **will** choose one of the P-/$–Bets

	DIE ROLL 1	DIE ROLL 2	DIE ROLL 3	DIE ROLL 4	DIE ROLL 5	DIE ROLL 6	DIE ROLL 7	DIE ROLL 8	DIE ROLL 9	DIE ROLL 10
LOTTERY A	175	175	175	0	0	0	0	0	0	0
LOTTERY B	175	170	165	0	0	0	0	0	0	0
LOTTERY C	170	170	170	0	0	0	0	0	0	0
LOTTERY D	175	175	0	0	0	0	0	0	0	0
LOTTERY E	110	110	110	110	110	110	110	0	0	0
LOTTERY F	110	105	100	110	105	100	110	0	0	0
LOTTERY G	105	105	105	105	105	105	105	0	0	0
LOTTERY H	110	110	110	110	110	110	0	0	0	0

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- 8 rounds: Algorithm choice—10' viewing time—10' decision/waiting time
- P-Bet vs $-Bet, e.g.: (110, 0.7; 0) vs (175, 0.3; 0)
- Generating complexity: 2. Scrambling rows

	DIE ROLL 1	DIE ROLL 2	DIE ROLL 3	DIE ROLL 4	DIE ROLL 5	DIE ROLL 6	DIE ROLL 7	DIE ROLL 8	DIE ROLL 9	DIE ROLL 10
LOTTERY A	175	170	165	0	0	0	0	0	0	0
LOTTERY B	110	105	100	110	105	100	110	0	0	0
LOTTERY C	170	170	170	0	0	0	0	0	0	0
LOTTERY D	105	105	105	105	105	105	105	0	0	0
LOTTERY E	175	175	175	0	0	0	0	0	0	0
LOTTERY F	110	110	110	110	110	110	110	0	0	0
LOTTERY G	110	110	110	110	110	110	0	0	0	0
LOTTERY H	175	175	0	0	0	0	0	0	0	0

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- 8 rounds: Algorithm choice—10' viewing time—10' decision/waiting time
- P-Bet vs \$-Bet, e.g.: (110, 0.7; 0) vs (175, 0.3; 0)
- Generating complexity: 3. shifting one category

	DIE ROLL 1	DIE ROLL 2	DIE ROLL 3	DIE ROLL 4	DIE ROLL 5	DIE ROLL 6	DIE ROLL 7	DIE ROLL 8	DIE ROLL 9	DIE ROLL 10
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LOTTERY E	175	175	175	0	0	0	0	0	0	0
LOTTERY F	0	0	110	110	110	110	110	110	110	0
LOTTERY G	0	0	110	110	110	110	110	110	0	0
LOTTERY H	175	175	0	0	0	0	0	0	0	0

Lottery-Selection Task

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- 8 rounds: Algorithm choice—10' viewing time—10' decision/waiting time
- P-Bet vs $-Bet, e.g.: (110, 0.7; 0) vs (175, 0.3; 0)
- Generating complexity: 4. Scrambling columns

	DIE ROLL 1	DIE ROLL 2	DIE ROLL 3	DIE ROLL 4	DIE ROLL 5	DIE ROLL 6	DIE ROLL 7	DIE ROLL 8	DIE ROLL 9	DIE ROLL 10
LOTTERY A	0	0	175	0	170	0	165	0	0	0
LOTTERY B	105	110	0	110	0	100	110	0	105	100
LOTTERY C	0	0	170	0	170	0	170	0	0	0
LOTTERY D	105	105	0	105	0	105	105	0	105	105
LOTTERY E	0	0	175	0	175	0	175	0	0	0
LOTTERY F	110	110	0	110	0	110	110	0	110	110
LOTTERY G	110	110	0	0	0	110	110	0	110	110
LOTTERY H	0	0	175	0	175	0	0	0	0	0

Feedback

- With self-reliance, subjects always know which lottery was selected
  ⇒ always give selection feedback
- Feedback manipulation: learning the lottery **outcome**, too

Feedback

- With self-reliance, subjects always know which lottery was selected
  $\Rightarrow$ always give selection feedback
- Feedback manipulation: learning the lottery **outcome**, too
- **NF**: reveal selected lottery $\rightarrow$ measure satisfaction
- **F**: reveal selected lottery + outcome $\rightarrow$ measure satisfaction

Hypotheses

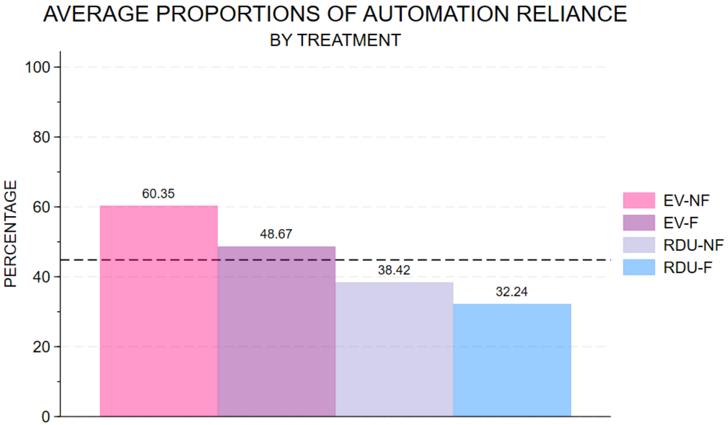
H1. Self-reliance sometimes results in the selection of dominated lotteries.

H2. Algorithm reliance differs between EV algorithm and RDU algorithm.

H3a. Algorithm reliance declines over time under outcome feedback.
H3b. "Win-stay–lose-shift."
H3c. "Lose-shift" is stronger for delegated choices.

Results

"Pre-H": Do participants exhibit an overall preference for self-reliance?



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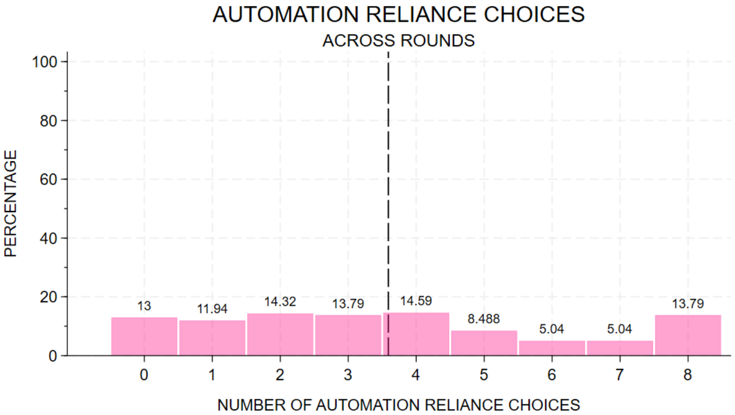

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H1. Self-reliance sometimes results in the selection of dominated lotteries.

→ Yes. *After round 2*, in around 35% of all choices.

⇒ There are potential gains from delegating.

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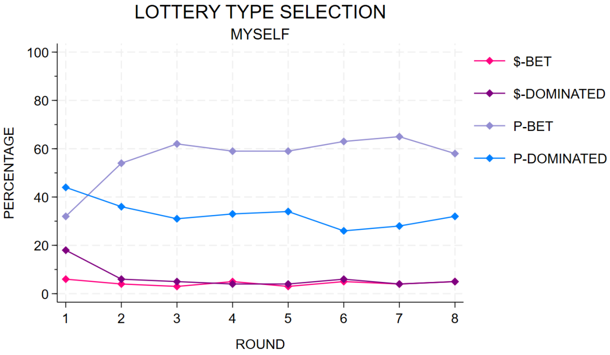

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Results

H2. Algorithm reliance differs between EV algorithm and RDU algorithm.

→ Yes. *Already in round 1*, by about 20 percentage points.

⇒ “Don’t Do What I Would Do, Do What’s Best!”

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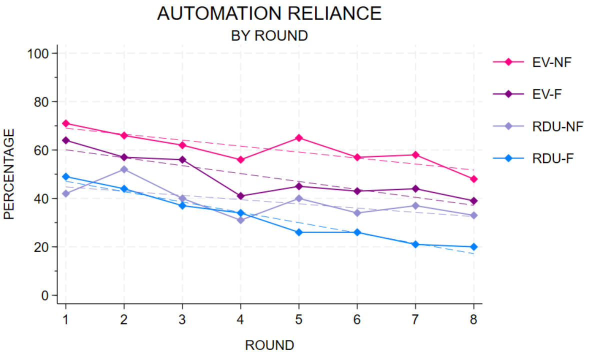

⇒ "Don't Do What I Would Do, Do What's Best!"

H3a. Algorithm reliance declines over time under outcome feedback.

→ Clearly so (and unsubstantially more than without).

Results

H3b. "Win-stay–lose-shift."

	Stay	Switch
EV-F		
Self-Reliance		
Win	84.23	15.77
Lose	59.09	40.91
Automation Reliance		
Win	79.75	20.25
Lose	61.45	38.55
RDU-F		
Self-Reliance		
Win	87.23	12.77
Lose	71.82	28.18
Automation Reliance		
Win	79.27	20.73
Lose	41.67	58.33

Note: Average frequencies of staying with or switching from one's automation or self-reliance decision after winning (positive feedback) or losing (zero feedback).

H3c. "Lose-shift" stronger after choosing an algorithm?

→ Yes. But exists only for RDU.

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Algorithm-Preference ‘Congruence’

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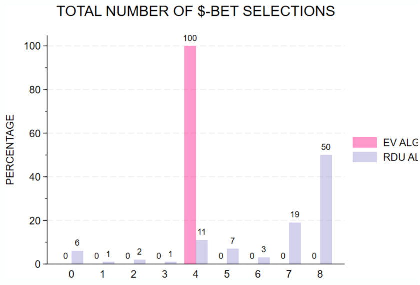
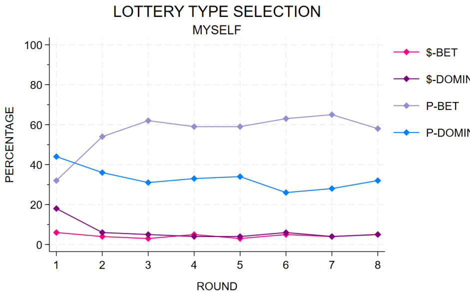
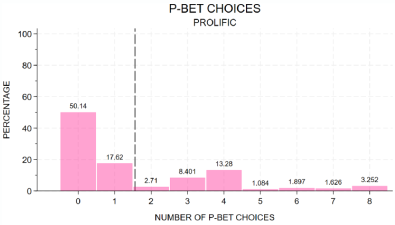
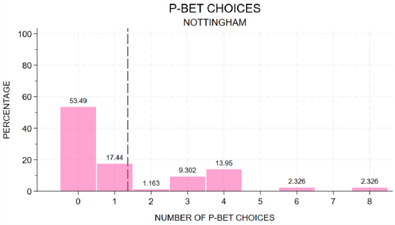




Weird first-part data?

- RDU-parameters at a first glance look somewhat different than those from Murad et al. (2016), but:





Simulation: what about the satisfied?

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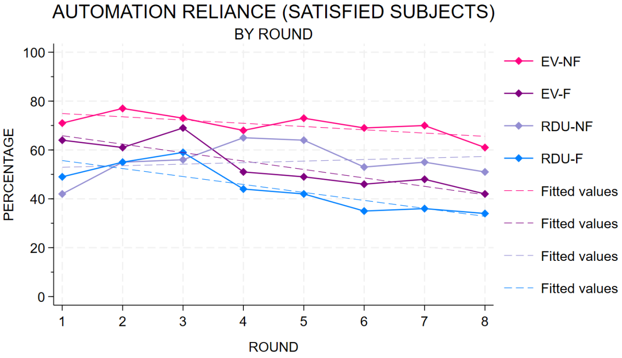
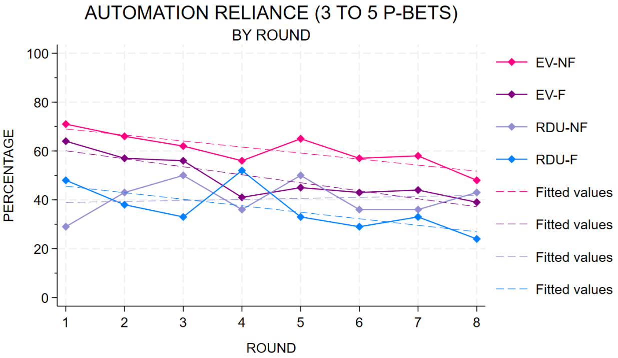


- downward trends weaker (except EV-F) but alive for all but RDU-NF
- relevant treatment comparisons continue to hold

Making EV and RDU comparable

- ...in terms of choice: using only RDU-obs. with 3-5 $-bet selections:



- ...also does not seem to change much (RDU slightly more attractive).

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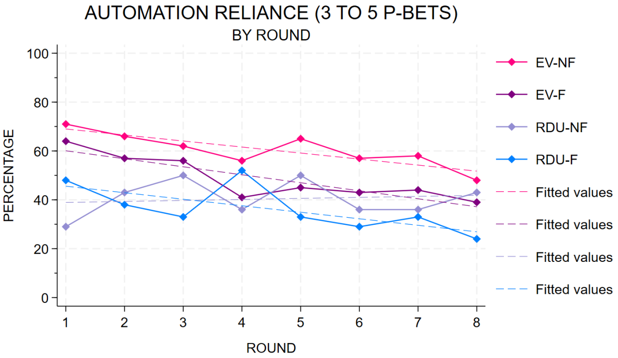

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Summing up

- We don't know too much about "AR where preferences matter"
- People (from the start) don't want an algorithm that tries to mimic their preferences
- ...but they do want one that maximises EV
- AR declines over time, somewhat more with outcome feedback
- Stronger reactions to 'algorithmic mistakes' only in RDU-F

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- ...but they do want one that maximises EV
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Interpretation:

- I know what I want, but does the machine? Does it add errors?

  → preferential algorithm → less people believe in its performance
- If I can 'commit to' a strategy that is 'better overall', I'll do it

  → commitment story → lower sensitivity

Thank You!